

Youngjae Cho

Machine Learning Researcher

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github.com/youngjae-cho | [Google Scholar](#)

Research interests: Robust and data-efficient machine learning, with an emphasis on preference optimization, active learning, Bayesian adaptation, and multimodal learning.

Education

KAIST | M.S. in Industrial & Systems Engineering

Mar 2022 - Feb 2024

Advisor: Il-Chul Moon

KAIST | B.S. in Industrial & Systems Engineering

Mar 2017 - Feb 2022

Publications

Learning Where It Matters: Geometric Anchoring for Robust Preference Alignment

Youngjae Cho, Jongsuk Kim, Ji-Hoon Kim

NeurIPS 2026 | Accepted | First author

Background-Aware Defect Generation for Robust Industrial Anomaly Detection

Youngjae Cho, Gwangyeol Kim, Sirojbek Safarov, Seongdeok Bang, Jaewoo Park

arXiv 2024 | Preprint | First author

Make Prompts Adaptable: Bayesian Modeling for Vision-Language Prompt Learning with Data-Dependent Prior

Youngjae Cho, HeeSun Bae, Seungjae Shin, YeoDong Youn, Weonyoung Joo, Il-Chul Moon

AAAI 2024 | First author

Sharpness-Aware Active Learning

Yoon-Yeong Kim*, Youngjae Cho*, JoonHo Jang, Byeonghu Na, Yeongmin Kim, Kyungwoo Song, Wanmo Kang, Il-Chul Moon

ICML 2023 | Co-first author

Improving Group-based Robustness and Calibration via Ordered Risk and Confidence Regularization

Seungjae Shin, Byeonghu Na, HeeSun Bae, JoonHo Jang, Hyemi Kim, Kyungwoo Song, Youngjae Cho, Il-Chul Moon

ICML 2022 Workshop on Spurious Correlations, Invariance and Stability

Predict Sequential Credit Card Delinquency with VaDE-Seq2Seq

Yeongmin Kim, Youngjae Cho, Hanbit Lee, Il-Chul Moon

IEEE SMC 2021

* Equal contribution.

Research Experience

Pyler | Research Scientist

Oct 2025 - Present

- Developed **GAPO**, a geometric anchoring method for robust preference optimization under noisy supervision (**NeurIPS 2026, accepted; first author**). Improved AlpacaEval 2.0 length-controlled win rate by 3.6 percentage points over SimPO.
- Built multimodal preference-optimization methods and training infrastructure for **Nemotron-Nano-12B-v2-VL** using SimPO and Megatron-Bridge; designed preference data for video temporal localization and content safety.
- Developed auditable LLM/VLM supervision by recovering decision trees from **299K reasoning traces**, improving content-safety macro F1 from **0.777 to 0.857**.

Aiv Co. | ML Research Scientist

Mar 2024 - Oct 2025

- Led research on background-aware diffusion for industrial anomaly detection (**arXiv 2024; first author**), using defect/background disentanglement to synthesize scarce anomalies.
- Evaluated generation quality on MVTec-AD and LOCO against DFMGAN and AnomalyDiffusion, then deployed the method to improve detector precision and recall on production data.

KAIST | Graduate Research

Mar 2022 - Feb 2024

- Developed Bayesian vision-language prompt learning with data-dependent priors for input-adaptive prompts under limited data and distribution shift (**APP, AAAI 2024; first author**).
- Developed a sharpness-aware acquisition strategy for active learning to improve generalization under a limited labeling budget (**SAAL, ICML 2023; co-first author**).

Academic Service

ICML 2026 | Gold Reviewer

NeurIPS 2026 | Reviewer

Awards

NVIDIA Nemotron Hackathon | Winner, Track B

2026

Team winner in Track B (Domain-Specialized Model). Built the preference-optimization stack and designed preference data for the winning video content-safety VLM.